

RESEARCH ARTICLE

Herb–drug interactions between androgenic Chinese herbal medicines and androgen receptor antagonist on tumor growth: Studies on two xenograft prostate cancer animal models

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Our previous study revealed that Epimedium Folium (EF) and Codonopsis Radix (CNR) significantly promoted tumor growth on a subcutaneous mouse model of prostate cancer (PCa) via enhancing the mRNA and protein expressions of androgen receptor (AR), while Astragalus Radix (AGR) inhibited tumor growth via suppressing the protein expression of AR. In the present study, we aimed to investigate the potential interactions between EF, CNR or AGR and AR antagonist (abiraterone acetate [ABI]) on the tumor growth using subcutaneous and orthotopic PCa mouse models. EF, CNR, AGR and ABI were intragastrically given to mice once every 2 days for 4 weeks. The pharmacokinetics of ABI were evaluated in the plasma of rats when combined with EF, CNR, or AGR. Our results demonstrated that EF or CNR could weaken the anti-tumor effects of ABI via increasing the AR expression involving activation of the PI3K/AKT and Rb/E2F pathways and decreasing the bioavailability of ABI, while AGR could enhance the anti-tumor effects of ABI through suppressing the AR expression via inhibiting the activations of PI3K/AKT and Rb/E2F pathways and increasing the bioavailability of ABI. These findings imply that cautions should be exercised when prescribing EF and CNR for PCa patients.

KEYWORDS

abiraterone acetate, androgen receptor, androgenic Chinese herbal medicines, pharmacokinetics, prostate cancer, tumor growth

Abbreviations: ABI, abiraterone acetate; ADT, androgen deprivation therapy; AGR, Astragalus Radix; AR, androgen receptor; BCA, bichloronic acid; CDK, cyclin-dependent kinases; CHM, Chinese herbal medicine; CLZ, Clearance; CNR, Codonopsis Radix; CRPC, castration-resistant prostate cancer; DAD, diode array detector; E2F1, E2 Transcription Factor 1; EF, Epimedium Folium; FBS, fetal bovine serum; IS, internal standard; PCa, prostate cancer; PI3K, phosphatidylinositol-4,5-bisphosphate 3-kinase; PIP2, phosphatidylinositol (4,5) biphosphate; PIP3, phosphatidylinositol (3,4,5) trisphosphate; PTEN, phosphatase and tensin homolog; PVDF, polyvinylidene fluoride; Rb, retinoblastoma; UPLC, ultraperformance liquid chromatography.

1 | INTRODUCTION

Prostate cancer (PCa) is the second most common malignancy in men in most regions of the world, with 1,276,106 new diagnosed cases and 358,989 deaths in 2018 (Bray et al., 2018). It has been well-known that androgens drive PCa progression through targeting related genes and signaling networks to promote PCa cell survival by

binding to the ligand binding domain of the androgen receptor (AR) during the development of PCa (Debes & Tindall, 2004; Tenakoon et al., 2014). It is crucial to inhibit androgen signaling either by depriving the tumor from androgens or by blocking the receptor activity. Therefore, androgen deprivation therapy (ADT) is currently the first-line treatment for advanced PCa in clinical practice (Gonçalves et al., 2018). Abiraterone acetate (ABI), a new generation of AR antagonist, could effectively block androgen biosynthesis and suppress AR expression in normal prostate glands and PCa. ABI has been approved for the treatment of metastatic castration-resistant prostate cancer (CRPC) (Goldberg & Berrios-Colon, 2013).

Many PCa patients, particularly those in Asian countries such as China, Japan, and Singapore, use Chinese herbal medicines (CHMs) as an adjuvant treatment while receiving Western therapeutic modalities (Singh et al., 2005). The use of CHMs in this group of cancer patients purportedly aims to alleviate the side effects induced by conventional chemotherapy and radiotherapy, enhance the body's general immunity, and improve the quality of life. However, it was reported that some of the CHMs commonly prescribed for the treatment of PCa were shown to contain phytoandrogens (Klempner & Bubley, 2012). This has raised concerns about the safety of using CHMs, especially those containing phytoandrogens, as an adjuvant therapy for PCa. Since phytoandrogens and androgen have many similar biological effects, especially in modulation of the AR, it is important to characterize whether CHMs containing phytoandrogens could counteract the effect of various ADT.

Our previous study screened 21 selected CHMs and found that 70% aqueous ethanol extract of *Epimedium Folium* (EF) and *Codonopsis Radix* (CNR) significantly promoted tumor growth via increasing the protein expression of AR in the tumor tissues, while *Astragalus Radix* (AGR) effectively suppressed the tumor growth via inhibiting the protein expression of AR in tumor tissues of 22Rv1 subcutaneous mouse model of PCa (Zhang et al., 2020). However, whether these three CHMs could influence the therapeutic effects of AR antagonists still remains unknown. Therefore, in the present study, we aimed to investigate the potential interaction between ABI and the three CHMs, including EF, CNR, and AGR on the tumor growth using subcutaneous and orthotopic animal models of PCa, and to illustrate the underlying molecular mechanisms. Moreover, the effects of these three CHMs on the oral pharmacokinetics of ABI were explored in the plasma of rats. The design of this study was followed the practical guidelines for phytopharmacological research (Heinrich et al., 2020; Izzo et al., 2020).

2 | MATERIALS AND METHODS

2.1 | Chemicals and reagents

ABI was purchased from Sigma-Aldrich (St. Louis, MO, USA). Physcion (Batch No.: A0045) was purchased from Chengdu Mansite Pharmaceutical Co. Ltd (Chengdu, Sichuan, China). The 70% aqueous ethanol extract of EF, CNR, and AGR was prepared as described in our previous study (Zhang et al., 2020). EF contained 0.73% icariin, while CNR had 0.52% lobetyolin. RPMI 1640 medium (Batch No.: A1049101), fetal

bovine serum (FBS) (Batch No.: 10270106), penicillin and streptomycin (Batch No.: 10378016) were obtained from Invitrogen (Carlsbad, CA, USA). Polyvinylidene fluoride (PVDF) membrane was purchased from the Bio-Rad (CA, USA). Acetonitrile and methanol were purchased from Duksan (Gyeonggi-do, South Korea). Other chemicals and reagents used in the current investigation were of analytical grade.

2.2 | Quality control of AGR

Quality control of AGR was performed using an ACQUITY UPLC system (Waters, USA) coupled with a diode array detector (DAD) according to the Chinese Pharmacopoeia (2015). Briefly, the extract of AGR was dissolved in 50% aqueous methanol (at a final concentration of 20 mg/mL) and injected into UPLC with autosampler. An Inertsil ODS-4 Analytical C18 Columns (4.6 × 250 mm, 5 μm, GL Sciences, USA) was used for the separation. The separation temperature was set at 30°C. The isocratic elution was performed with the mixture of water and acetonitrile (68:32, v/v) at a flow rate of 1 mL/min. The detected wavelength was 201 nm and the sample injection volume was 10 μL. AGR was found to contain 0.043% astragaloside IV. The HPLC chromatogram of AGR was shown in Figure 1A.

2.3 | Cell culture

22Rv1 cells were purchased from the American Type Culture Collection (ATCC, Manassas, VA, United States), and maintained in RPMI 1640 medium supplemented with 10% fetal bovine serum (FBS), 100 U/mL penicillin, and 100 mg/mL streptomycin (Gibco, New York, USA) in a 5% CO₂ humidified incubator at 37°C. The passages 10 to 15 of 22Rv1 cells were used to establish the subcutaneous and orthotopic PCa animal models.

2.4 | Animals

Male nude mice (6–8-week-old) and male Sprague–Dawley (SD) rats (200–220 g) were obtained from the Laboratory Animal Services Centre, The Chinese University of Hong Kong (CUHK). All animals were housed under regulated temperature (23 ± 2°C) and humidity (50 ± 10%) and maintained on a 12-h light–dark cycle. They were allowed to access to standard diet and clean water ad libitum. The animal experiment was approved by the Animal Experimentation Ethics Committee of CUHK (Reference No.: 19-032-HMR).

2.5 | Establishment of the subcutaneous PCa animal model and drug treatment

The subcutaneous animal model was established using 22Rv1 cells. Briefly, 1×10^6 22Rv1 cells were re-suspended in 0.1 mL serum-free medium and diluted with Matrigel (1:1) and subcutaneously injected

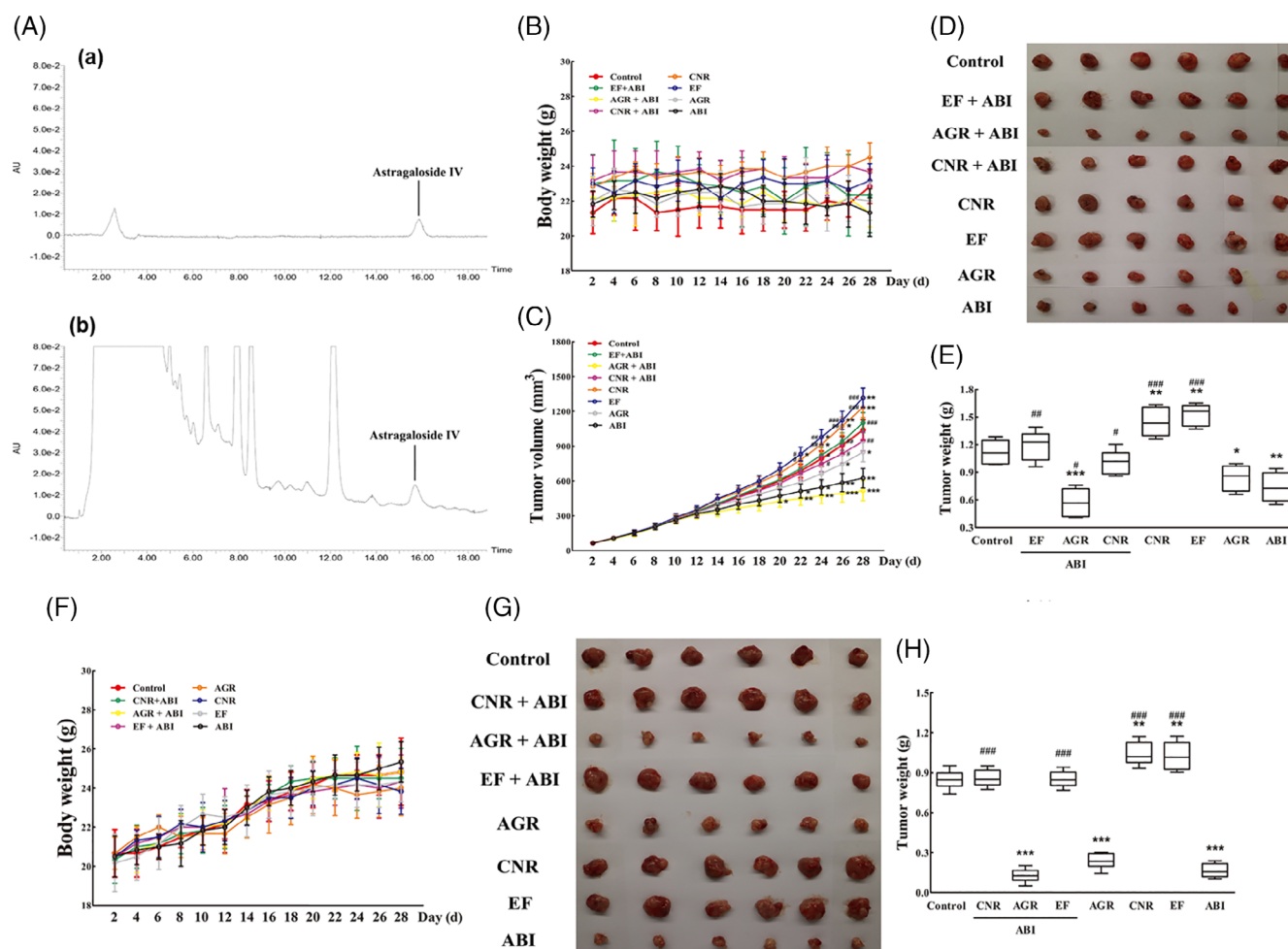


FIGURE 1 Effects of the combination of EF, CNR, or AGR and ABI on the body weight, tumor volume and tumor weight in the subcutaneous and orthotopic PCa animal models. (A) HPLC chromatograph of astragaloside IV (i) and AGR (ii). (B) Mouse body weight of 22Rv1 subcutaneous animal model treated with the corresponding drugs for a period of 4 weeks was presented. (C) Tumor volume of 22Rv1 subcutaneous animal model treated with the corresponding drugs for a period of 4 weeks was presented. (D) Photo of 22Rv1 subcutaneous tumor tissues after the mice were sacrificed. (E) Tumor weight of the nude mice after the mice were sacrificed. (F) The body weight of 22Rv1 orthotopic PCa mice treated with the corresponding drugs for a period of 4 weeks was presented. (G) Photos of 22Rv1 orthotopic tumor tissues after the mice were sacrificed. (H) Tumor weight of the nude mice after sacrifice. Data were expressed as means $\pm$ SEM ($n = 8$). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ as compared to the control group. # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ as compared to the ABI group [Colour figure can be viewed at wileyonlinelibrary.com]

into the right flank of each nude mouse at 4–6 weeks of age using a 25-gauge needle. The tumor volume was measured on alternate days using the modified ellipsoid formula (Volume = length $\times$ width² $\times$ 0.5) (Wu et al., 2017). When the tumors were palpable (more than 50 mm³), usually after a week of implantation, a total of 64 mice were randomized into eight testing groups ($n = 8$ per group): (a) the control group (0.5% CMC-Na), (b) ABI group (20 mg/kg), (c) EF group (161.7 mg/kg), (d) CNR group (345.6 mg/kg), (e) AGR group (37.1 mg/kg), (f) ABI (20 mg/kg) plus EF (161.7 mg/kg) group, (g) ABI (20 mg/kg) plus CNR (345.6 mg/kg) group, and (h) ABI (20 mg/kg) plus AGR (37.1 mg/kg) group. The selection of EF, CNR, AGR, and ABI doses and treatment durations was based on our previous study (Zhang et al., 2020). ABI, EF, CNR, and AGR were suspended in 0.5% CMC-Na and intragastrically (i.g.) given to mice once every 2 days for

4 consecutive weeks. During the drug treatment, body weights and tumor volumes of the mice were also monitored every 2 days for 4 weeks. One hour after the last dosing, the mice were sacrificed and tumor tissues harvested, weighed, and photographed by two observers who were blinded to the animal grouping. Tumor tissues were stored at -80°C for further biochemical experiments such as quantitative real-time PCR (qRT-PCR) and Western blot analysis.

2.6 | Establishment of the orthotopic PCa animal model and drug treatment

The orthotopic animal model was also established by using 22Rv1 cells as previously described with some modification (Tuomela

et al., 2009). Briefly, a low midline abdominal incision (5–10 mm) was made through the abdominal musculature, followed by carefully moving the bladder and seminal vesicles to expose the prostate. Then 5×10^5 22Rv1 cells in 20 μ L volume of serum-free medium was injected into the prostate of the nude mouse at 4–6 weeks of age using a 29-gauge needle. After 2 weeks of implantation, a total of 64 mice were randomized into eight groups of eight animals each. The grouping, doses of the corresponding drugs, and treatment schedules were similar to those described in the subcutaneous animal model experiment. All drugs were i.g. given to mice once every 2 days for four consecutive weeks. During the drug treatment, the body weights of the mice were monitored every 2 days for 4 weeks. One hour after the last dosing, the mice were sacrificed and tumor tissues harvested, weighed, and photographed by two observers who were blinded to the animal grouping. Tumor tissues were stored at -80°C for further biochemical experiments such as qRT-PCR and Western blot analysis.

2.7 | qRT-PCR analysis

Total RNA was isolated from the tumor tissues with TRIzol Reagent (Invitrogen, Carlsbad, CA, USA). The concentration of RNA was determined spectrophotometrically at 260 nm. The quality of RNA was estimated by the ratio of absorbance at 260 and 280 nm. The values of A260/A280 from 1.9 to 2.1 were considered as acceptable. Total RNA was reverse-transcribed with the High Capacity cDNA RT Kit (Applied Biosystems, Foster City, CA, USA) according to the manufacturer's instructions. qRT-PCR was carried out using the SYBR Premix Ex Taq (Tli RNaseH Plus) kit (Takara Bio Inc., Chuo-ku, Tokyo, Japan) on the step-one Fast Real-Time PCR system and mouse TaqMan gene expression assays (Applied Biosystems, assay ID: AR, Mm00442688_m1 and β -actin, Mm02619580_g1). The relative expression mRNA of AR was normalized to the amount of β -actin in the same cDNA using the relative quantification method ($2^{-\Delta\Delta\text{CT}}$ method).

2.8 | Western blot analysis

Tumor tissues were homogenized in cold RIPA buffer containing protease inhibitor cocktail and phosphatase inhibitors. Protein concentrations of tumor tissues were measured using the bicinchoninic acid (BCA) protein reagent (Pierce Chemical, Rockford, IL, USA). Equal amounts of protein of different samples were then separated on a SDS-polyacrylamide gel and then transferred to a PVDF membrane. After blocking with 5% nonfat milk for 1 h, the membranes were probed with primary antibodies against anti-AR (catalog no.: #5153), anti-PI3K (p110 α) (catalog no.: #4249), anti-AKT (catalog no.: #4691), anti-p-AKT (Ser473) (catalog no.: #4060), anti-PTEN (catalog no.: #9188), anti-FOXO1 (catalog no.: #2880), anti-p-FOXO1 (Ser256) (catalog no.: #84192), anti-Rb (catalog no.: #9313), anti-p-Rb (Ser807/811) (catalog no.: #8516), anti-E2F1 (catalog no.: #3742), anti-Cyclin D1 (catalog no.: #55506), anti-CDK4 (catalog no.: #12790), anti-CDK6 (catalog no.: #3136), and anti- β -actin (catalog no.: #4970) (Cell Signaling Technology,

MA, USA) overnight at 4°C . Then HRP-conjugated secondary antibody was used at room temperature for 1 h. Finally after washing the membranes, the protein bands were detected by the ECL western blotting detection reagents (Life Technologies, CA, USA). The quantification of western blotting results was done using Image J software (National Institutes of Health, Bethesda, MD, USA).

2.9 | Pharmacokinetic study

2.9.1 | Drug treatment and blood sample collection

Male SD rats (weighing 200–220 g) were randomized into 4 groups ($n = 6$ per group): (a) ABI group (20 mg/kg), (b) ABI (20 mg/kg) plus EF group (161.7 mg/kg), (c) ABI (20 mg/kg) plus CNR group (345.6 mg/kg), and (d) ABI (20 mg/kg) plus AGR group (37.1 mg/kg). ABI, EF, CNR, and AGR were suspended in 0.5% CMC-Na and i.g. given to the rats once every 2 days for 4 weeks. After the last dosing, the rats were anesthetized with isoflurane and the blood samples collected from orbital venous plexus of rats via heparinized capillary tubes at 10, 20, 30, 60, 120, 240, 360, 480, and 1,440 min. Plasma was obtained after centrifugation at $7,000 \times g$ for 15 min at 4°C and frozen at -80°C for further ultraperformance liquid chromatography (UPLC) analysis.

2.9.2 | Analytical conditions

The concentration of ABI in plasma was measured using an Acquity UPLC (Waters, USA) coupled with a diode array detector (DAD). The chromatographic separation was achieved on a Spherisorb ODS1 column (C18 Column) (4×150 mm, $5 \mu\text{m}$, Waters, USA), which was maintained at room temperature. The mobile phase was composed of acetonitrile (solvent A) and 12 mM ammonium acetate in water (solvent B). The elution progress was as follow: 80 to 85% solvent A in 0 to 7.5 min, 85 to 95% solvent A in 7.5 to 10 min, and 95% solvent A in 10 to 12 min. The flow rate was at 1 mL/min. The separation was conducted at room temperature and the detection wavelength at 255 nm.

Physcion was used as the internal standard (IS). The 12-point standard curve (1.5, 3, 6, 12, 18, 24, 30, 36, 42, 48, 54, and 60 $\mu\text{g/mL}$) was constructed by plotting the peak area ratio of ABI and IS against the nominal concentration of the calibration standards in control rat plasma. The results were considered to fit linear regression analysis using a $1/x^2$ (x = concentration) weighting factor. The calibration curve needed to show a good linear correlation ($r^2 \geq 0.99$) between the peak area ratio and the concentrations.

2.10 | Pharmacokinetic and statistical analysis

The noncompartmental pharmacokinetic model was used to determine the pharmacokinetic parameters by using the DAS 2.0 (Drug and

statistics) software. The area under the curve ($AUC_{0-\infty}$) meant that the area under the plasma ABI concentration-time curves from zero to infinity and was determined by the trapezoidal rules (Yang et al., 2012). Peak plasma ABI concentration (C_{max}) and the time when it occurred (T_{max}), as well as the $T_{1/2z}$ (the time for the terminal plasma ABI concentration to drop down by half) could be deduced from the curves. Clearance (CL_Z) was determined by the equation $(F \times \text{Dose}) / AUC_{0-\infty}$, where F is the fraction of the dose available in the systemic circulation. Data were expressed as means $\pm$ SEM. Multiple group comparisons were carried out using a one-way analysis of variance (ANOVA) followed by Dunnett's test. SPSS software version 20.0 (Chicago, IL, USA) was used to perform the statistical analysis. Statistically significant was considered when $p < 0.05$.

3 | RESULTS

3.1 | Effects of the combination of EF, CNR, or AGR and ABI on the body weight, tumor volume, and tumor weight in the subcutaneous PCa animal model

As shown in Figure 1, there was no substantial alteration on the body weight of mice after administration of all drugs (Figure 1B). Treatment with AGR or ABI alone markedly reduced the tumor volume from 22nd and 26th day onward (after 22nd day: $p < 0.05$, 24th day: $p < 0.05$, 26th day: $p < 0.01$ for AGR and after 26th day: $p < 0.01$ for ABI) (Figure 1C) and tumor weight ($p < 0.05$ and $p < 0.01$, respectively) (Figure 1E), respectively, as compared with the control group. However, after administration with EF or CNR, the tumor volume was markedly increased from the 22nd day onward (22nd day: $p < 0.05$ for EF, 24th day: $p < 0.05$ for both, after 24th day: $p < 0.01$ for both), when compared with the control group (Figure 1C). Similarly, the tumor weight was also drastically increased in the EF and CNR groups ($p < 0.01$ for both), when compared with the control group (Figure 1E).

On the other hand, treatment with the combination of EF and ABI or CNR with ABI did not show significant effects on tumor volume and tumor weight, as compared with the control group (Figure 1C,E). When compared with the ABI group, the combination of EF and ABI or CNR with ABI drastically accelerated the tumor growth (24th day: $p < 0.05$ for both, 26th day: $p < 0.05$ and $p < 0.01$, respectively, 28th day: $p < 0.01$ and $p < 0.001$, respectively) (Figure 1C) and increased the tumor weight ($p < 0.05$ and $p < 0.01$, respectively) (Figure 1E). In contrast, the combination of AGR and ABI treatment effectively inhibited the tumor volumes from the 20th day onwards (20th day: $p < 0.05$, 22nd to 24th day: $p < 0.01$, after 26th day: $p < 0.001$) (Figure 1C), and the tumor weight, when compared with the control group ($p < 0.001$) (Figure 1E). Moreover, the combination of AGR and ABI treatment was more effective in suppressing the tumor weight ($p < 0.05$), as compared to the ABI group. These results amply demonstrated that both EF and CNR could weaken the anti-tumor effect of ABI on tumor growth in the subcutaneous PCa animal model, while AGR and ABI may act synergistically to suppress the tumor growth of PCa.

3.2 | Effects of the combination of EF, CNR, or AGR and ABI on the body weight, tumor volume, and tumor weight in the orthotopic PCa animal model

As shown in Figure 1, treatment with all drugs did not show any marked effects on the body weight of the mice (Figure 1F). The tumor weight markedly decreased after treatment with AGR or ABI alone ($p < 0.001$ for both), as compared with the control group. On the contrary, administration with EF or CNR significantly accentuated the tumor weight of the mice ($p < 0.01$ for both), when compared with the control group. No significant influence in tumor weight was found in the orthotopic PCa animal model after administration with the combination of EF and ABI or CNR and ABI, when compared with the control group. The combination of EF and ABI or CNR and ABI substantially increased the tumor weight ($p < 0.001$ for both), as compared to the ABI group (Figure 1H). In contrast, treatment with the combination of AGR and ABI substantially reduced the tumor weight ($p < 0.001$), as compared with the control group. These results were similar to that in the subcutaneous PCa animal model, indicating that both EF and CNR could adversely influence the anti-tumor effects of ABI and promote the tumor growth in these two xenograft PCa animal models. In contrast, the combination of AGR and ABI treatment was effective in attenuating the tumor growth in PCa, suggesting a synergism between AGR and ABI in the anti-PCa effect.

3.3 | Effects of the combination of EF, CNR, or AGR and ABI on the mRNA and protein expressions of AR in the subcutaneous PCa animal model

The effects of the combination of EF, CNR, or AGR and ABI on AR expressions in the tumor tissues of the subcutaneous PCa animal model were shown in Figure 2. Administration with EF or CNR alone drastically enhanced the mRNA ($p < 0.001$ for both) and protein ($p < 0.05$ for both) expressions of AR, as compared with the control group. However, treatment with AGR or ABI alone could inhibit the AR protein expression ($p < 0.05$ and $p < 0.01$, respectively), as compared with the control group. Although no marked difference was found in the protein expression of AR between the combination of EF and ABI or CNR and ABI treatment groups, (Figure 2A), the mRNA expression of AR was substantially increased after the combination of EF and ABI or CNR and ABI treatment ($p < 0.05$ and $p < 0.01$, respectively) (Figure 2B), as compared with the control group. The combination of EF and ABI or CNR and ABI significantly up-regulated the mRNA ($p < 0.05$ and $p < 0.01$, respectively) and protein ($p < 0.01$ for both) expressions of AR, when compared with the ABI group. In contrast, the combination treatment of AGR and ABI resulted in marked attenuation of the mRNA ($p < 0.05$) and protein ($p < 0.01$) expressions of AR, when compared with the control group. Treatment with ABI alone significantly inhibited the mRNA expressions of AR ($p < 0.05$), when compared with the control group. The results revealed that the combined treatment of EF or CNR with ABI could induce tumor growth by enhancing the AR expression, while combined treatment

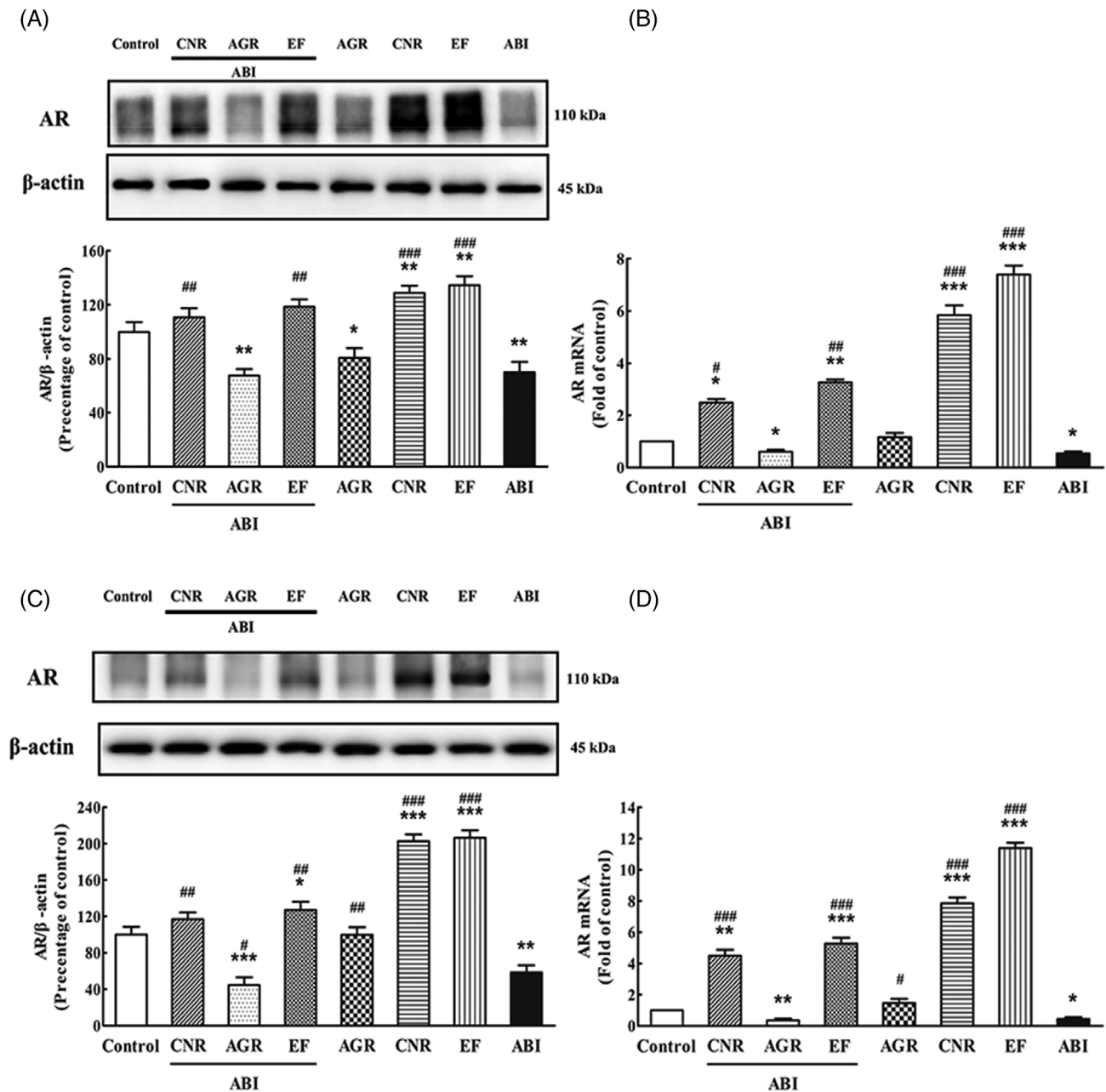


FIGURE 2 Effects of the combination of EF, CNR, or AGR and ABI on the protein and mRNA expressions of AR in the subcutaneous and orthotopic PCa animal models. Data were expressed as means $\pm$ SEM ($n = 3-6$). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ as compared to the control group. # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ as compared to the ABI group

between AGR and ABI exerted an anti-cancer effect through inhibition of AR expression.

3.4 | Effects of the combination of EF, CNR, or AGR and ABI on the mRNA and protein expressions of AR in the orthotopic PCa animal model

As shown in Figure 2, treatment with EF, CNR, or ABI alone markedly enhanced the mRNA ($p < 0.001$ for both) and protein

($p < 0.001$ for both) expressions of AR in the tumor tissues of the mice, as compared with the control group. However, treatment with AGR or ABI alone significantly suppressed the mRNA ($p < 0.05$ for both) and protein ($p < 0.01$ for both) expressions of AR, when compared with the control group. On the other hand, the combination treatment of EF and ABI dramatically increased the mRNA ($p < 0.001$) and protein ($p < 0.05$) expressions of AR, as compared with the control group. Similarly, the combination treatment of CNR and ABI also markedly enhanced the mRNA expression of AR in the tumor tissues of orthotopic PCa animal model ($p < 0.01$), as com-

pared to the control group (Figure 2C,D). Most importantly, when compared with the ABI group, co-treatment with EF and ABI or CNR and ABI substantially enhanced the mRNA ($p < 0.001$ for both) and protein ($p < 0.01$ for both) expressions of AR. However, the combination of AGR and ABI significantly suppressed the mRNA ($p < 0.01$) and protein ($p < 0.001$) expressions of AR, as compared with the control group. The combination of AGR and ABI more effectively inhibited the protein expression of AR ($p < 0.05$), as compared with ABI group. All these findings conspicuously indicated that EF or CNR treatment could mitigate the anti-tumor effects of ABI via enhancing the AR expression. In contrast, AGR may be a potential anti-cancer herb that could synergize with the anti-tumor effects of ABI in PCa.

3.5 | Effects of the combination of EF, CNR, or AGR and ABI on the PI3K/AKT signaling pathway in the subcutaneous PCa animal model

We further investigated the molecular mechanisms of the combination of EF, CNR or AGR and ABI in the subcutaneous PCa animal model. As illustrated in Figure 3, treatment with EF or CNR alone drastically increased the protein expressions of PI3K ($p < 0.001$ for both), p-AKT/AKT ($p < 0.001$ for both), and p-FOXO1/FOXO1 ($p < 0.001$ for both), but decreased the protein expression of PTEN ($p < 0.001$ for both) in the tumor tissues of the nude mice, as compared to the control group. In contrast, AGR or ABI treatment alone could markedly enhance the expression of PTEN ($p < 0.001$ for both),

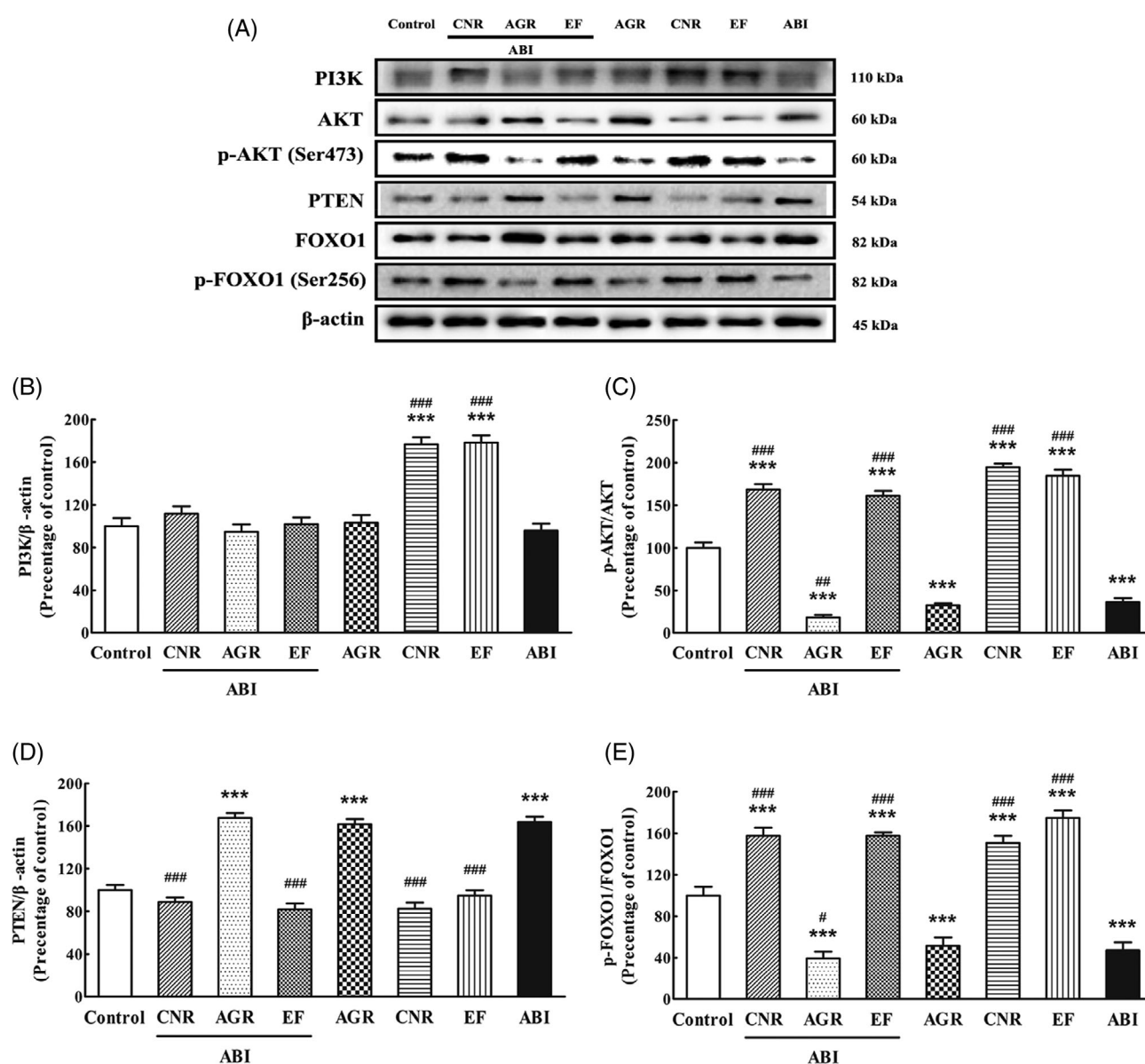


FIGURE 3 Effects of the combination of EF, CNR, or AGR and ABI on the protein expression of PI3K (B), p-AKT/AKT (C), PTEN (D) and p-FOXO1/FOXO1 (E) in the subcutaneous PCa animal model. Data were expressed as means $\pm$ SEM ($n = 3$). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ as compared to the control group. # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ as compared to the ABI group

but suppress the expression of p-AKT/AKT ($p < 0.001$ for both) and p-FOXO1/FOXO1 ($p < 0.001$ for both), as compared with the control group. On the other hand, the combination of EF and ABI or CNR and ABI exponentially increased the protein expressions of p-AKT/AKT ($p < 0.001$ for both) and p-FOXO1/FOXO1 ($p < 0.001$ for both), but exerted no significant effects on the protein expressions of PI3K and PTEN in the tumor tissues of the nude mice, when compared to the control group (Figure 3B,D). However, the co-treatment of EF and ABI or CNR and ABI could substantially attenuate PTEN ($p < 0.001$ for both) protein expressions and up-regulate the p-AKT ($p < 0.001$ for both) and p-FOXO1 ($p < 0.001$ for both) protein expressions in the tumor tissues of the nude mice, when compared with the ABI group. In addition, the combination of AGR and ABI treatment led to the up-regulated protein expression of PTEN ($p < 0.001$), while decreased the

ratios of p-AKT/AKT ($p < 0.001$) and p-FOXO1/FOXO1 ($p < 0.001$) in the tumor tissues of nude mice, as compared to the control group. The combination of AGR and ABI also showed significant suppressive effects on the ratios of p-AKT/AKT ($p < 0.01$) and p-FOXO1/FOXO1 ($p < 0.05$), when compared to the ABI group.

3.6 | Effects of the combination of EF, CNR, or AGR and ABI on the Rb/E2F signaling pathway in the subcutaneous PCa animal model

As shown in Figure 4, treatment with EF or CNR alone substantially down-regulated the protein expression of Rb and up-regulated the protein expressions of p-Rb ($p < 0.001$ for both), E2F1 ($p < 0.01$ for

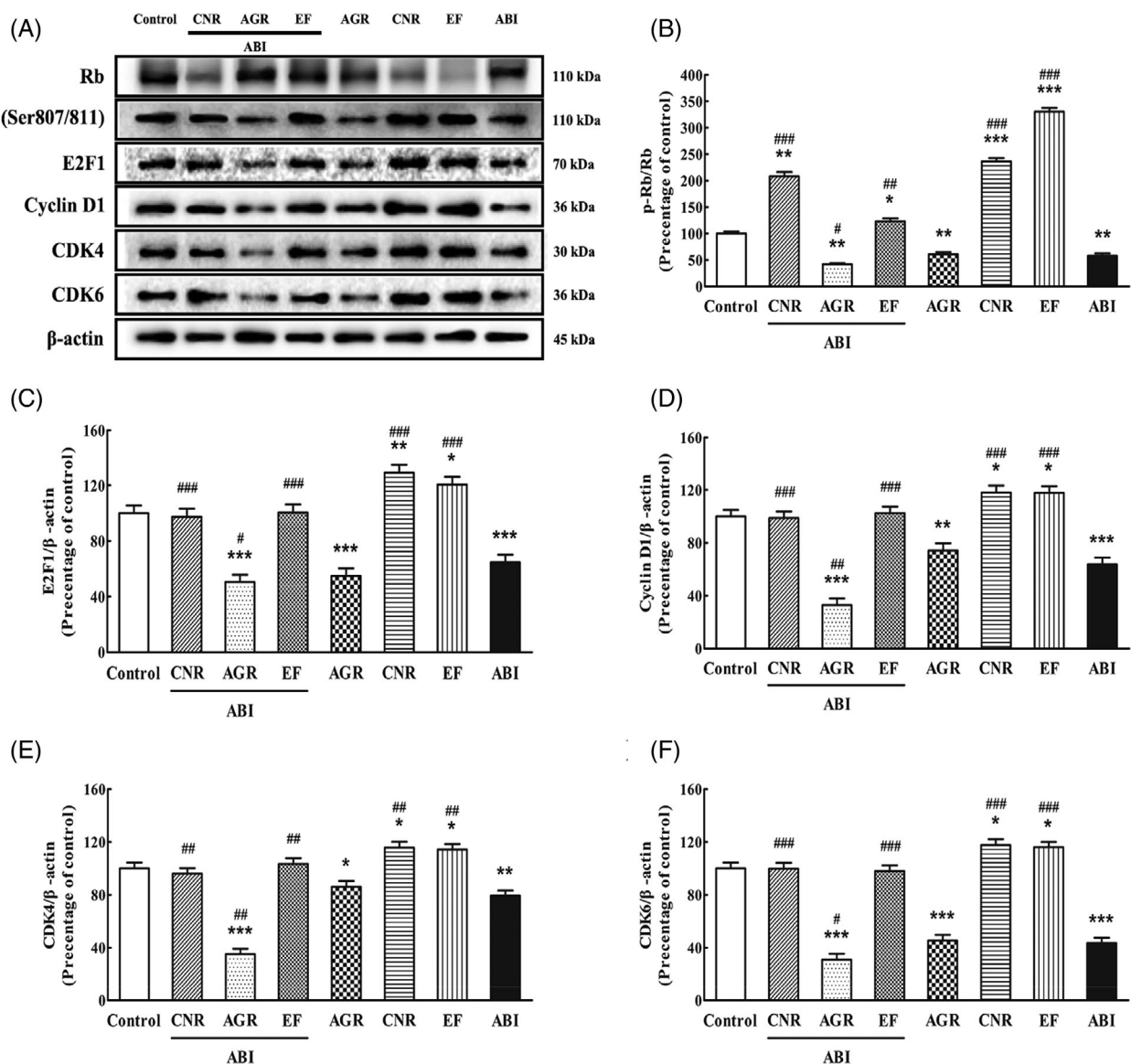


FIGURE 4 Effects of the combination of EF, CNR, or AGR and ABI on the protein expression of p-Rb/Rb (B), E2F1 (C), cyclin D1 (D), CDK4 (E), and CDK6 (F) in the subcutaneous PCa animal model. Data were expressed as means $\pm$ SEM ($n = 3$). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ as compared to the control group. # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ as compared to the ABI group

CNR, $p < 0.05$ for EF), Cyclin D1 ($p < 0.05$ for both), CDK4 ($p < 0.05$ for both), and CDK6 ($p < 0.05$ for both), as compared with the control group. In contrast, treatment with AGR or ABI alone effectively inhibited the protein expressions of p-Rb/Rb ($p < 0.01$ for both), E2F1 ($p < 0.001$ for both), Cyclin D1 ($p < 0.01$ and $p < 0.001$, respectively),

CDK4 ($p < 0.05$ and $p < 0.01$, respectively), and CDK6 ($p < 0.001$ for both), when compared with the control group. On the other hand, the combination of CNR and ABI or EF and ABI significantly induced the phosphorylation of Rb ($p < 0.01$ and $p < 0.05$, respectively), while the protein expressions of E2F1, cyclin D1, CDK4, and CDK6 showed

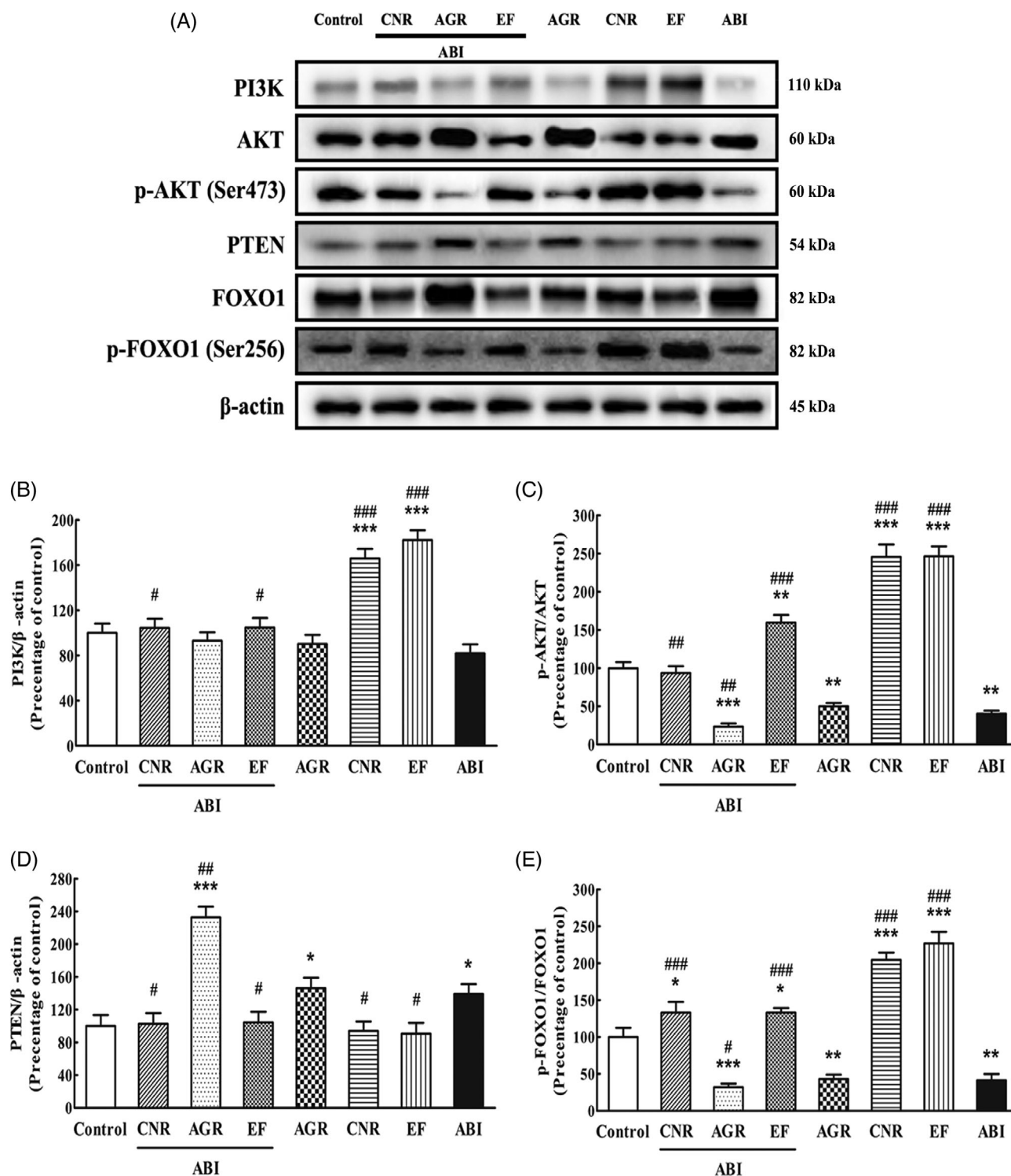


FIGURE 5 Effects of the combination of EF, CNR, or AGR and ABI on the protein expression of PI3K (B), p-AKT/AKT (C), PTEN (D), and p-FOXO1/FOXO1 (E) in the orthotopic PCa animal model. Data were expressed as means $\pm$ SEM ($n = 3$). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ as compared to the control group. # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ as compared to the ABI group

no significant differences, as compared to the control group. However, when compared with the ABI group, the combination of CNR and ABI or EF and ABI drastically activated the Rb/E2F signaling by suppressing the protein expression of Rb ($p < 0.001$ and $p < 0.01$, respectively) and enhancing the protein expressions of E2F1 ($p < 0.001$ for both), Cyclin D1 ($p < 0.001$ for both), CDK4 ($p < 0.01$ for both) and CDK6 ($p < 0.001$ for both). In contrast, the combination treatment of AGR and ABI led to decreased protein expressions of p-Rb/Rb ($p < 0.01$), E2F1 ($p < 0.001$), Cyclin D1 ($p < 0.001$), CDK4 ($p < 0.001$), and CDK6 ($p < 0.001$) in tumor tissues of nude mice, as compared with the control group. In particular, the combination treatment of AGR and ABI was effective in inhibiting the Rb/E2F signaling by increasing the ratio of p-Rb/Rb ($p < 0.01$) and suppressing the E2F1 ($p < 0.05$), Cyclin D1 ($p < 0.01$), CDK4 ($p < 0.01$), and CDK6 ($p < 0.05$) protein expressions, as compared to the ABI treatment group.

3.7 | Effects of the combination of EF, CNR, or AGR and ABI on the PI3K/AKT signaling pathway in the orthotopic PCa animal model

To illustrate the molecular mechanisms of the combination of EF, CNR, or AGR and ABI in the orthotopic PCa animal model, we also determined the involvement of the PI3K/AKT signaling pathway. As shown in Figure 5, administration with EF or CNR alone substantially increased the protein expressions of PI3K ($p < 0.001$ for both), p-AKT/AKT ($p < 0.001$ for both), and p-FOXO1/FOXO1 ($p < 0.001$ for both), but decreased the protein expression of PTEN ($p < 0.05$ for both) in the tumor tissues, as compared with the control group. In contrast, treatment with AGR or ABI alone significantly up-regulated the protein expression of PTEN ($p < 0.05$ for both), but markedly down-regulated the ratios of p-AKT/AKT ($p < 0.01$ for both) and p-FOXO1/FOXO1 ($p < 0.01$ for both) in the tumor tissues, as compared with the control group. On the other hand, the combination of EF and ABI exponentially enhanced the ratios of p-AKT/AKT ($p < 0.01$) and p-FOXO1/FOXO1 ($p < 0.05$), and the combination of CNR and ABI also markedly up-regulated the ratio of p-FOXO1/FOXO1 ($p < 0.05$), as compared to the control group. No significant differences were found in the protein expressions of PI3K and PTEN of the combination of EF and ABI or CNR and ABI, as compared with the control group. On the other hand, when compared with the ABI group, the combination of EF and ABI or CNR and ABI could dramatically elevate the protein expressions of PI3K ($p < 0.05$ for both), p-AKT/AKT ($p < 0.01$ and $p < 0.001$, respectively), and p-FOXO1/FOXO1 ($p < 0.001$ for both), but reduce the protein expression of PTEN ($p < 0.05$ for both). On the contrary, the combination of AGR and ABI effectively inhibited the ratios of p-AKT/AKT ($p < 0.001$) and p-FOXO1/FOXO1 ($p < 0.001$), while enhanced the protein expression of PTEN ($p < 0.001$), as compared with the control group. Moreover, the combination of AGR and ABI also significantly up-regulated the protein expression of PTEN ($p < 0.01$), but down-regulated the ratios of p-AKT/AKT ($p < 0.01$) and p-FOXO1/FOXO1 ($p < 0.05$), as compared to the ABI group.

3.8 | Effects of the combination of EF, CNR, or AGR and ABI on the Rb/E2F signaling pathway in the orthotopic PCa animal model

The effects of combination of EF, CNR, or AGR and ABI on the Rb/E2F signaling pathway in the orthotopic PCa animal model were shown in Figure 6. After treatment with EF or CNR alone, the p-Rb/Rb ($p < 0.001$ for both), E2F1 ($p < 0.001$ for both), Cyclin D1 ($p < 0.001$ for both), CDK4 ($p < 0.01$ for both), and CDK6 ($p < 0.05$ for both) protein expressions were substantially increased, as compared with the control group. However, administration with AGR or ABI alone inhibited the protein expressions of p-Rb/Rb ($p < 0.01$ for both), E2F1 ($p < 0.001$ and $p < 0.01$, respectively), Cyclin D1 ($p < 0.001$ and $p > 0.05$, respectively), CDK4 ($p < 0.001$ and $p < 0.01$, respectively), and CDK6 ($p > 0.05$ and $p < 0.001$, respectively), when compared with the control group. In contrast, the combination of CNR and ABI or EF and ABI dramatically elevated the ratio of p-Rb/Rb ($p < 0.01$ and $p < 0.05$, respectively), when compared with the control group. In particular, the combination of CNR and ABI or EF and ABI significantly increased the protein expressions of p-Rb/Rb ($p < 0.001$ and $p < 0.01$, respectively), E2F1 ($p < 0.01$ for both), Cyclin D1 ($p < 0.05$ for both), CDK4 ($p < 0.01$ for both), and CDK6 ($p < 0.001$ for both), when compared with the ABI group. Conversely, the combination of AGR and ABI effectively inhibited the protein expressions of p-Rb/Rb ($p < 0.01$), E2F1 ($p < 0.001$), cyclin D1 ($p < 0.001$), CDK4 ($p < 0.001$), and CDK6 ($p < 0.001$), as compared with the control group. Meanwhile, the combination of AGR and ABI significantly inhibited the protein expressions of E2F1 ($p < 0.01$), Cyclin D1 ($p < 0.001$), CDK4 ($p < 0.05$), and CDK6 ($p < 0.05$), as compared with the ABI group.

3.9 | Effects of EF, CNR, or AGR on the concentration of ABI in plasma of rats

Effects of EF, CNR, or AGR on the multiple-dose pharmacokinetics of ABI were assessed by determining the concentrations of ABI in the plasma of rats. The plasma ABI concentration-time curves were constructed using 12-point calibration standards (range 1.5–60 $\mu\text{g/mL}$) in the rat plasma. Of which the standard curve possessed a reliable reproducibility over the concentrations of standard curve solutions across the calibration range. As shown in Figure 7A, the standard curve was prepared by measuring the best fit of the peak-area ratios (peak area of ABI/peak area of IS) versus concentration ratios with fitting to $y = 0.2082x - 0.0605$ using the weighting factor ($1/x^2$). The average regression ($n = 6$) was found to be ≥ 0.992 .

As shown in Figure 7B, after combination treatment of EF or CNR and ABI, the concentrations of ABI in the plasma of the rats were exponentially attenuated, as compared to the ABI treatment alone. However, treatment with the combination of AGR and ABI significantly accentuated the concentration of ABI in the plasma of the rats, when compared with the ABI group. The pharmacokinetic parameters of the ABI group and co-treatment groups were shown in Table 1. The

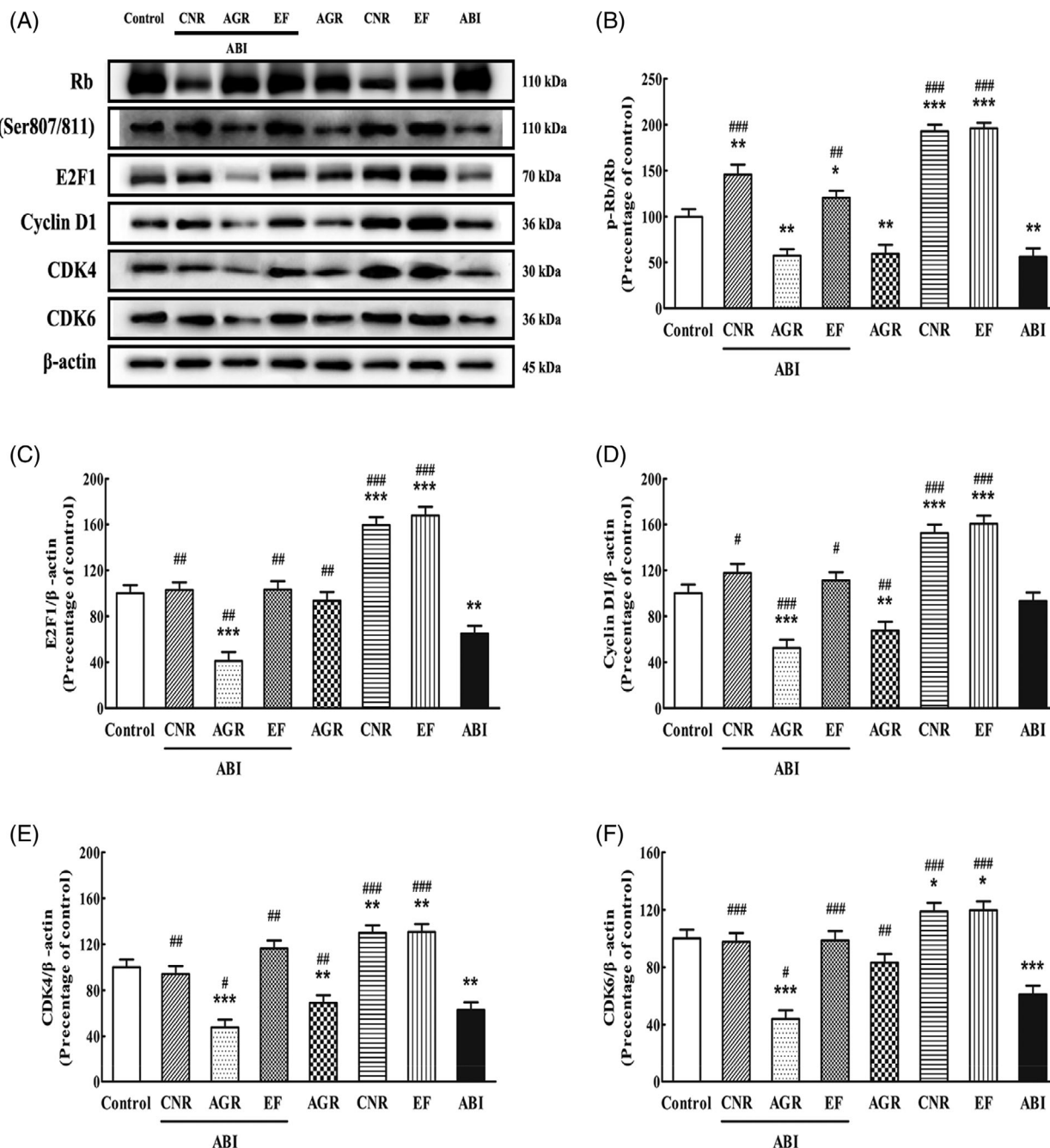


FIGURE 6 Effects of the combination of EF, CNR, or AGR and ABI on the protein expression of p-Rb/Rb (B), E2F1 (C), cyclin D1 (D), CDK4 (E), and CDK6 (F) in the orthotopic PCa animal model. Data were expressed as means $\pm$ SEM ($n = 3$). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ as compared to the control group. # $p < 0.05$, ## $p < 0.01$, and ### $p < 0.001$ as compared to the ABI group

AUC_{0-∞} of ABI in the combination of EF and ABI or CNR and ABI drastically decreased ($p < 0.01$ for both), as compared with the ABI-treated group. In contrast, the AUC_{0-∞} of ABI in the combination of AGR and ABI has shown a 25.2% increase ($p < 0.05$), as compared with the ABI-treated group. Meanwhile, the maximum concentration of ABI (C_{max}) in the plasma of ABI treatment group was 1.49 ± 0.11 $\mu\text{g/mL}$, while the ABI C_{max} in the combination of the EF and ABI-treated rats substantially reduced by 40.1% ($p < 0.01$), as compared with the ABI-

treated group. However, the AGR and ABI co-treated rats displayed significantly higher C_{max} (by 49.8%, $p < 0.01$), as compared with the ABI treatment group. In addition, the T_{max} value of the combination of CNR and ABI was significantly decreased (by 76.7%, $p < 0.001$), as compared with the ABI control. On the other hand, the combination of EF and ABI or CNR and ABI displayed significantly higher CL_z/F value than the ABI control, with increases of 53.6% ($p < 0.01$) and 48.2% ($p < 0.01$), respectively, as compared with the ABI control.

FIGURE 7 Effects of EF, CNR, or AGR on the concentration of ABI in the plasma of rats. (A) The standard curve of ABI/IS was prepared by measuring the best fit of the peak-area ratios (peak area of ABI/peak area of IS) versus concentration (ABI concentration/IS concentration) ratios. (B) Mean plasma concentration versus time profile of ABI in the rat plasma following oral administration of ABI alone or together with EF, CNR, or AGR to rats. Data were expressed as means $\pm$ SEM ($n = 6$) [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

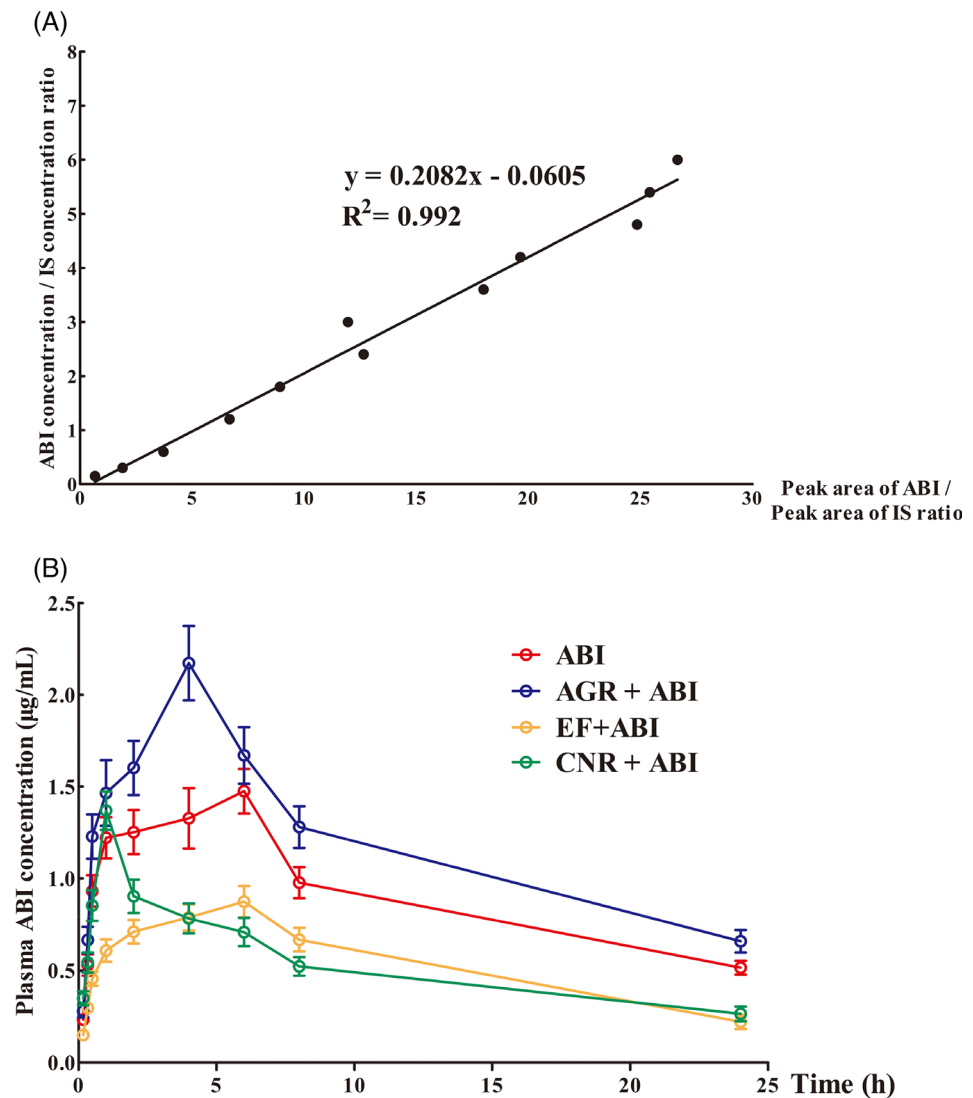


TABLE 1 Pharmacokinetic parameters of ABI (20 mg/kg) after multiple doses

Parameter	ABI (20 mg/kg)	ABI (20 mg/kg)		
		EF (161.7 mg/kg)	CNR (345.6 mg/kg)	AGR (37.1 mg/kg)
AUC _{0-∞} (μg·h/mL)	33.2 $\pm$ 0.9	16.3 $\pm$ 0.7**	17.4 $\pm$ 1.1**	41.9 $\pm$ 2.8*
T _{1/2z} (h)	15.4 $\pm$ 0.7	10.7 $\pm$ 1.2	12.5 $\pm$ 0.71	14.5 $\pm$ 2.0
T _{max} (h)	5.5 $\pm$ 0.4	5.1 $\pm$ 0.5	1.3 $\pm$ 0.23***	4.7 $\pm$ 0.5
CLz/F (L/h/kg)	0.6 $\pm$ 0.03	1.2 $\pm$ 0.05**	1.1 $\pm$ 0.03**	0.5 $\pm$ 0.04
C _{max} (μg/mL)	1.5 $\pm$ 0.1	0.9 $\pm$ 0.1**	1.4 $\pm$ 0.1	2.2 $\pm$ 0.2**

Note: Data were expressed as means $\pm$ SEM ($n = 6$). * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ as compared to the ABI group.

4 | DISCUSSION

In the present study, we selected EF, CNR, and AGR for investigating the potential interaction between androgenic CHMs and ABI on the tumor growth using two xenograft PCa animal models. The results demonstrated that treatment with ABI alone significantly inhibited the tumor growth and AR expressions on these two xenograft PCa animal

models. However, the combination of EF or CNR with ABI significantly increased tumor growth and AR expression in tumor tissues in both PCa mouse models, as compared to the ABI-treated group. These findings suggest that both EF and CNR could undermine the anti-tumor effects of ABI. The mechanisms underlying the observed antagonism involve with the promotion of the protein expression of AR and enhancement of the activation of PI3K/Akt and Rb/E2F pathways. On

the other hand, the combination of AGR and ABI exerted more potent anti-tumor effect as it suppressed tumor volume, tumor weight and protein expression of AR in tumor tissues in the two xenografts PCa mouse models, and the underlying mechanisms are associated with inhibiting the activation of PI3K/Akt and Rb/E2F pathways. Our findings unambiguously indicate that AGR acts synergistically with ABI to exert more potent anti-tumor effect on PCa.

It has been recently reported that appropriately 40% of primary PCa and 70% of CRPC genomic alterations could result in the activation of the PI3K/AKT signaling pathway. The crosstalk and reciprocal feedback mechanism between PI3K/AKT signaling pathway and AR signaling were also found in many studies (Carver et al., 2011). The experimental evidence revealed that the PI3K/AKT signaling pathway played a critical role in the AR-mediated PCa progression. The activation of PI3K could induce the phosphorylation of phosphatidylinositol (4,5) bisphosphate (PIP2) into phosphatidylinositol (3,4,5) trisphosphate (PIP3), which directly phosphorylates the AKT signaling. The activation of AKT signaling then phosphorylates the downstream effector FOXO1, an inhibitor that could directly bind to the AR in a ligand-independent manner (Bohrer et al., 2013), and subsequently induce the translocation of FOXO1 from the nucleus to the cytoplasm, thereby favoring transactivation of AR (Liu, Li, Gan, Kao, & Huang, 2008). Particularly, the activation of AKT signaling also depends on the tumor suppressor PTEN, which acts as a PIP3 phosphatase to antagonize the PI3K pathway, and further suppresses the AKT signaling (Chalhoub & Baker, 2009). Therefore, inhibition of the PI3K/AKT signaling pathway would decrease the AR protein levels and slow down the progression of PCa.

In our two xenograft PCa animal models, administration of ABI significantly suppressed the tumor growth via inhibition of the protein expression of AR and the activation of PI3K/AKT pathway. However, the combination of CNR and ABI, or EF and ABI, could markedly enhance tumor growth, AR protein expression and the activation of PI3K/AKT, as compared with the ABI alone treatment group. These results suggest that CNR or EF could interfere with the effects of ABI via PI3K/AKT signaling pathway. In contrast, AGR combined with ABI effectively inhibited the activation of PI3K/AKT signaling, and even exhibited better inhibitory effects on tumor growth than ABI alone treatment, indicating that AGR and ABI interact synergistically in inhibiting the progression of PCa.

As the downstream mediator of AR, the Rb pathway also plays crucial roles during cell proliferation and tumor growth in the carcinogenesis and progression of PCa. It is well-known that cell proliferation is generally induced by high Cyclin/CDK levels, especially the formation of Cyclin D1/CDK4 or CDK6, and the increased phosphorylation of Rb. The activation of the formation of cyclin D1/CDK4 or CDK6 also directly correlates with the activation of AR (Taneja, Ha, & Garabedian, 2001). Hypophosphorylation of Rb protein was reported to be associated with the attenuation of AR in both ADPC and AIPC (Yuan et al., 2006). In addition, previous studies have also shown that overexpression of E2F1 could bypass ADT and enhance cell proliferation in an androgen-sensitive LNCaP cell line (Libertini et al., 2006), indicating that E2F was also a critical mediator in the Rb pathway in

cell cycle progression in PCa. Most importantly, E2F1 was demonstrated to repress both the mRNA and protein expression of AR by suppressing the AR promoter activity in PCa (Davis et al., 2006). All these findings revealed that Rb/E2F signaling pathway is closely associated with the AR-mediated PCa development. Our results demonstrated that treatment with ABI alone dramatically inhibited the phosphorylation of Rb and protein expression of E2F1, as well as the cell cycle-related protein expressions in the two xenograft PCa animal models. However, the combination of CNR or EF with ABI obviously enhanced the protein expressions of p-Rb/Rb, E2F1, Cyclin D1, CDK4, and CDK6, as compared with the ABI-treated group, indicating that CNR or EF could adversely interfere the anti-tumor effects of ABI via reducing the inhibitory effects of ABI on Rb/E2F signaling pathway. On the other hand, the combination of AGR and ABI effectively suppressed the phosphorylation of Rb and inhibited the cell cycle-related protein such as Cyclin D1, CDK4, and CDK6, suggesting that AGR treatment could augment the inhibitory effects of ABI on Rb/E2F signaling pathway.

Laboratory and clinical studies of herb-drug interactions have clearly shown that the therapeutic efficacy and toxicity of pharmaceutical drugs could be negatively affected by combination treatment with CHMs (Kennedy & Seely, 2010). Therefore, it is critical to investigate the potential pharmacokinetic interaction between CHMs and oral pharmaceuticals in *in vivo* models. In the present study, we investigated, for the first time, the interaction between the androgenic CHMs and ABI in rats. The results demonstrated that the co-administration of EF and ABI significantly decreased the plasma ABI concentration, as compared with the ABI treatment alone group, indicating that EF could significantly reduce the therapeutic effect of ABI in PCa by decreasing the plasma concentrations of ABI. In addition, although the C_{max} was not dramatically altered after co-treatment with CNR and ABI, as compared with the ABI-treated group, the T_{max} value was significantly reduced by co-treatment with CNR and ABI, indicating that CNR could attenuate the anti-tumor effects of ABI by decreasing the bioavailability of ABI treatment. On the other hand, the rats co-treated with AGR and ABI showed significantly augmented effects on ABI pharmacokinetics such as the $AUC_{0-\infty}$ and C_{max} , when compared with the ABI treatment group. The experimental observation suggests that AGR could markedly increase the bioavailability of ABI to enhance the anti-tumor therapeutic effects of ABI in PCa.

Although the detailed mechanisms underlying the herb-drug interactions of EF, CNR or AGR and ABI require further elucidated, their phytochemical components are considered to play important roles in the interaction effects. Icariin, a major bioactive constituent isolated from EF, stimulated the release of nitric oxide via modulating the AR and PI3K/Akt pathways (Koizumi, Yu, Hashimoto, Ouchi, & Okabe, 2010). Saponins of CNR have been reported to stimulate the AR expression (Zhou et al., 2016). However, the polysaccharides of AGR have been reported to possess growth inhibitory effects on various cancers such as lung cancer (Park & Park, 2018), breast cancer (Liu et al., 2019), cervical cancer, and leukemia (Zhou et al., 2018). Moreover, the polysaccharides of AGR have been reported to synergistically enhance the anti-tumor efficacy of cisplatin in cervical

cancer (SiHa) cells (Liu et al., 2019). These chemical ingredients of EF, CNR, or AGR may be partially responsible for the interaction with ABI. More studies are needed to identify which compounds or fractions of these CHMs play the critical role in the herb–drug interactions among the Chinese herbs and AR antagonists for PCa.

5 | CONCLUSION

In our present study, we have demonstrated that androgenic CHMs, such as EF and CNR, could promote PCa development and suppress the anti-tumor effects of ABI. The effects were, at least in part, related to the activation of PI3K/Akt and Rb/E2F pathways, which resulted in increased AR expression. On the contrary, AGR and ABI synergistically exerted anti-tumor effects on PCa via inhibiting the protein expression of AR through suppressing the activations of PI3K/Akt and Rb/E2F pathways. These findings provide experimental evidence to support clinical recommendations, that is, PCa patients should not be prescribed with EF and CNR. PCa patients on AR antagonists should avoid the consumption of any CHMs with androgenic property. In addition, we should pay particular attention to ensure safety and treatment efficacy when combining CHMs and anti-PCa pharmaceuticals.

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CONFLICT OF INTEREST

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

AUTHORS' CONTRIBUTIONS

Yan-Fang Xian and Zhi-Xiu Lin conceived the research idea and designed the experimental protocols. Zhen-Biao Zhang performed the in vivo experiments. Zhen-Biao Zhang, Yan-Feng Huang, and Dan-Dan Luo collected the experimental data. Zhen Hu and Siu-Po Ip conducted UPLC analysis. William Chi Shing Cho and Chi Fai Ng performed the data analysis. Zhen-Biao Zhang drafted the manuscript. William Chi Shing Cho, Chi Fai Ng, Yan-Fang Xian, and Zhi-Xiu Lin revised the manuscript.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

All experimental procedures were approved by the Animal Experimentation Ethics Committee of The Chinese University of Hong Kong (Ref. No. 19-032-HMR).

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